

A variable damping viscous damper for control of buildings under wind loading

Mao Li^a, Mingmei Shi^b, Weiqing Fu^{b,*}

^a Department of Civil & Environmental Engineering, University of Alberta, Edmonton, T6H2G7, Canada

^b School of Civil Engineering, Qingdao University of Technology, Qingdao, 266520, China

ARTICLE INFO

Keywords:

Variable damping
Viscous damper
Passive control
Wind-induced vibration control
Benchmark model

ABSTRACT

High-rise buildings are susceptible to low-frequency vibrations induced by wind loading, and the traditional constant damping viscous damper (CDVD) falls short in effectively suppressing vibrations of high-rise buildings under wind loadings because of its constant damping coefficient. To address this, a novel variable damping viscous damper (VDVD) is first designed and fabricated. The characterization tests were conducted to evaluate the variable damping characteristics under different velocities. The control performance of the VDVD in high-rise buildings under wind loadings with varying wind pressures is simulated and analysed, as comparisons with CDVD. The test results show that the damping coefficient of the VDVD increases with the input velocity. Furthermore, it exhibits the capability to achieve both constant and variable damping through the adjustment of the pre-pressure force in the spring. Compared with cases using CDVD, the VDVD demonstrates a significantly greater reduction in structural response, with cumulative dissipated energy ratios ranging from 1.0 to 3.14 as wind pressure increases. The VDVD offers excellent potential for various engineering applications in terms of new concepts and new devices for vibration control and mitigation induced by wind loadings.

1. Introduction

As a result of urbanization in the past decades, the high-rise building is a prevalent choice in densely populated regions for accommodating residents. However, high-rise buildings are inherently vulnerable to wind-induced vibrations due to their low natural frequencies [1]. Damping is an efficient way for high-rise buildings to mitigate the influence of wind loadings and achieve optimal performance under such conditions. Unfortunately, the intrinsic damping in high-rise structures decreases as building height increases [2], and it is further affected by material degradation over time [3,4]. Therefore, enhancing intrinsic damping through enhancements in material strength and diameter of structure members is neither practical nor effective. Adding damping to the primary structures becomes a well-accepted engineering practice in the field [5,6]. Various damping devices, including semi-active control systems such as magnetorheological (MR) dampers and electrorheological (ER) dampers, as well as passive control mechanisms like constant damping viscous dampers and non-linear viscous dampers, have been proposed and employed in wind-induced vibration control [7].

Nonetheless, studies have indicated that, regardless of the type of control device, the control performance is closely related to the damping ratio of the structure [8,9]. It has been proved that increasing the external damping coefficient enhances seismic control performance in buildings [10]. Various researchers have developed different kinds of damping devices with high damping coefficients

* Corresponding author.

E-mail addresses: mao13@ualberta.ca (M. Li), mingmeishi1212@163.com (M. Shi), fuweiqing@qut.edu.cn (W. Fu).

[11–15]. Undoubtedly, control devices equipped with high damping characteristics are adept at generating higher damping forces and dissipating greater energy during extreme loading scenarios. However, such dampers operate like extremely rigid components in non-extreme situations, offering minimal hysteretic effects. The displacement is restricted as rigidity increases, but this comes at the cost of amplifying the acceleration and internal forces in the damper and main structure. Stated differently, while high damping devices may demonstrate excellent performance in decreasing displacements, they amplify the effects of abrupt impacts on the structures. Also, high damping damper may increase the acceleration response of the structure which has been noticed in paper [10]. In this regard, the viscous damper with constant damping or high damping may not always meet the requirement for high-rise buildings to achieve optimal performance under diverse wind loadings. As such, variable damping is a desirable feature for complex wind loadings in high-rise buildings [16,17].

Efforts have been made to develop variable damping devices in the community, and one of the representative examples is the development of a semi-active control system with adjustable damping, MR dampers and ER dampers [18]. Several researchers have conducted studies to investigate the control performance of those semi-active control systems. For instance, Su et al. [19] conducted studies in wind-induced vibration control of a feed supporting system using ER dampers. Bharti et al. [20] utilize MR dampers to minimize the damage between two adjacent tall buildings during seismic events. These studies collectively demonstrate that the semi-active control devices can offer additional damping for high-rise buildings, leading to improved control performance. However, there is no doubt that semi-active control devices require less energy supplies than active control, they still necessitate a constant power supply, which may not be available during extreme events, and the reliability of the sensor network remains a concern, especially for long service life. As such, passive systems with alterable damping are more favorable due to their simplicity and reliability.

To address this, significant efforts have been made in developing passive control devices. For instance, Hazaveh et al. [21] introduced a novel design of a displacement-dependent viscous damper capable of producing variable damping force at specific displacements. Zhao et al. [22] developed a rotational hydraulic damper with a variable damping coefficient that fluctuates and non-linearly increases with the rotational angle of the valve and the angular speed of the rotor. Xu et al. [23] presented and tested a variable-damping-coefficient viscous fluid damper based on the annular gap viscous damper. Additionally, the nonlinear viscous damper has also been widely used in wind or earthquake-induced vibration control [24–28]. While existing non-linear viscous dampers can provide a variable damping coefficient and damping force, those tend to decrease with the input velocity, resulting in a smaller force at high velocities and a decreased ability for energy dissipation under high input velocities. This poses a potential threat to the safety of structures subjected to high-intensity wind or earthquakes.

To address this challenge, this study aimed to provide a variable damping coefficient and damping force within the context of passive control and free from external energy supplies, following the framework shown in Fig. 1. Firstly, the variable damping viscous damper equipped with the property of increasing damping coefficient with input velocities is designed and fabricated, as detailed in Section 2. Subsequently, Section 3 presents the characterization test scheme and results of VDVD, wherein the hysteretic curves, maximum forces, and energy dissipation of VDVD are analysed. Before the wind-induced vibration control performance evaluation of VDVD, the numerical model of VDVD and CDVD are introduced and verified against the test results. Information on the building model, wind loadings, and the analysis method is also briefly provided in Section 4. The analysis results of VDVD in high-rise buildings for wind-induced vibration control, including VDVD with and without the constant damping stage, are presented in Section 5. The results of this research hold excellent potential for various engineering applications in terms of new concepts and new devices for vibration control and mitigation induced by wind loadings.

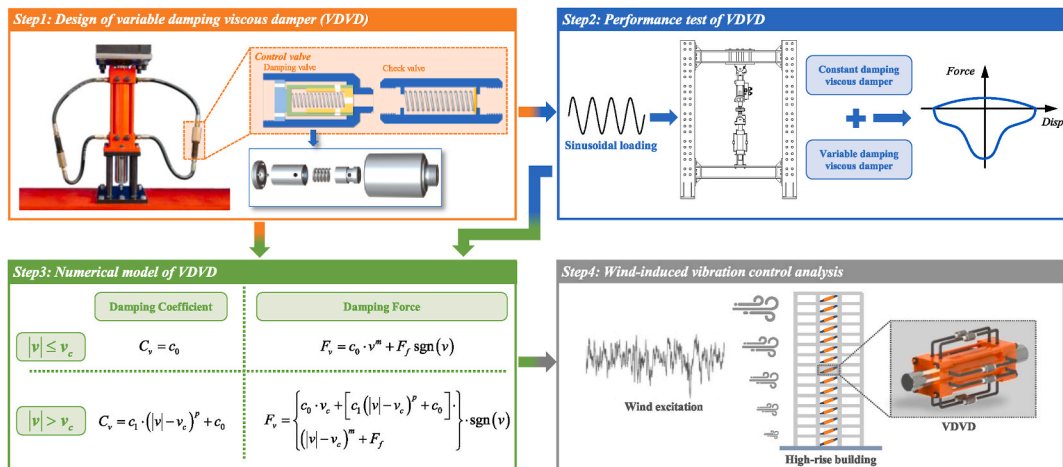


Fig. 1. Research framework flowchart.

2. Design and working principle of VDVD

In this section, the design philosophy of VDVD is presented. The detailed working principle of the control valve, the core component of VDVD, is introduced as well as that of the whole damper.

2.1. Design of VDVD

As shown in Figs. 2 and 3, the VDVD mainly consists of a double-ended damper and external pipes. The damper includes the cylinder, piston, and piston rod. The control valves are arranged within the external pipe on both sides of the cylinder in contrasting directions. Instead of using internal orifices on the piston as CDVD, the oil flows through the external pipe and control valve driven by the pressure caused by the motion of the piston. The piston has a clearance fit with the cylinder to endow smooth motion of the piston and ideal isolation of the hydraulic oil in both chambers. Two pairs of high-pressure hoses are adopted as the external pipe for easy installation and replacement of damping valves. In real engineering applications, control valves can be embedded in the piston for better integrality.

2.2. Working principle of control valves

The control valve is the core part of the VDVD, which consists of the damping valve and the check valve. The function of the damping valve is to modulate the damping coefficient, while the role of the check valve is to govern the direction of oil flow. Two valves are connected in series by an adapter nut, as shown in Fig. 4. In this regard, the oil can only flow from the damping valve to the check valve, precluding any reverse flow, thereby ensuring the intended direction of oil flow.

As shown in Figs. 5 and 6, the damping valve consists of a valve body, a movable disk, a fixed disk, a spring, and a support nut. The spring is placed inside the movable disk and fixed disk, the movable disk is placed inside the fixed disk, and the fixed disk is fixed in the valve body through the support nut. The movable disk and the fixed disk are in close contact, and hence the liquid cannot flow through the gap between the two disks but only through the inflow orifice and the damping orifices and outflow orifices (the red dash line in Fig. 5). Several damping orifices are set along the central axis near the top of the fixed disk and set at the same place in the movable disk.

As shown in Fig. 5, the oil flows as the red dotted line through the inflow orifices, the damping orifice, the gap between the fixed disk and valve body and the outflow orifice in the support nut. Depending on the stiffness and length of the spring, the damping valve has two different working scenarios.

(1) Spring with pre-compression.

When the length of the spring is greater than the space between the movable disk and the fixed disk, the spring will be compressed resulting in a pre-pressure force. Since the area of the inflow orifice is smaller than the external pipe, there is a pressure difference between the inside and outside of the movable orifices. When the pressure from oil is less than the pre-pressure of the spring, the movable disk will not move inside as the resultant force of the movable disk is in the same direction as the pre-pressure force of the spring. As such, the damping orifice keeps the original area, and the VDVD behaves as the characteristic of CDVD (stage 1 in Fig. 7 (a)). As the velocity of external excitation increases, once the pressure around the inflow orifice surpasses the pre-pressure force of the spring, the spring will be compressed, and the area of damping orifices will decrease due to the movement of the movable disk (stage 2 in Fig. 7 (a)). As a result, the damping coefficient increases due to the decrease in the area of damping orifices. Eventually, the damping orifice will close at a threshold velocity, and the VDVD will perform as a rigid body (stage 3 in Fig. 7 (a)). Alternatively, the length of the movable disk can also be increased. Therefore, damping orifices remain in a specific small area, and the VDVD will keep a fixed damping coefficient no matter how the input velocity increases when the left end of the movable disk contacts the inside bottom of the fixed disk (See Fig. 7 (b)).

(2) Spring without pre-pressure.

When the length of the spring is equal to that between the movable disk and the fixed disk, the spring will not have the pre-pressure force. Therefore, the spring will be compressed once the oil flows into the damping valve, and the damping coefficient of the VDVD will increase as the area of damping orifices decreases immediately (Fig. 7 (c)). The damping coefficient will vary in real-time with the input velocity at the beginning of loading.

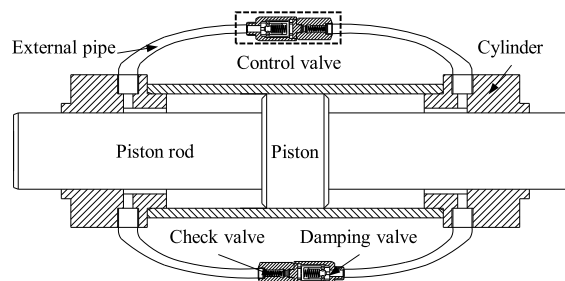


Fig. 2. Illustration of VDVD.

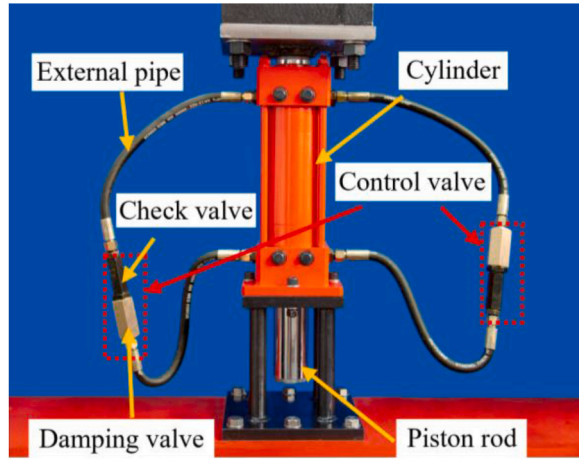


Fig. 3. The photo of VDVD.

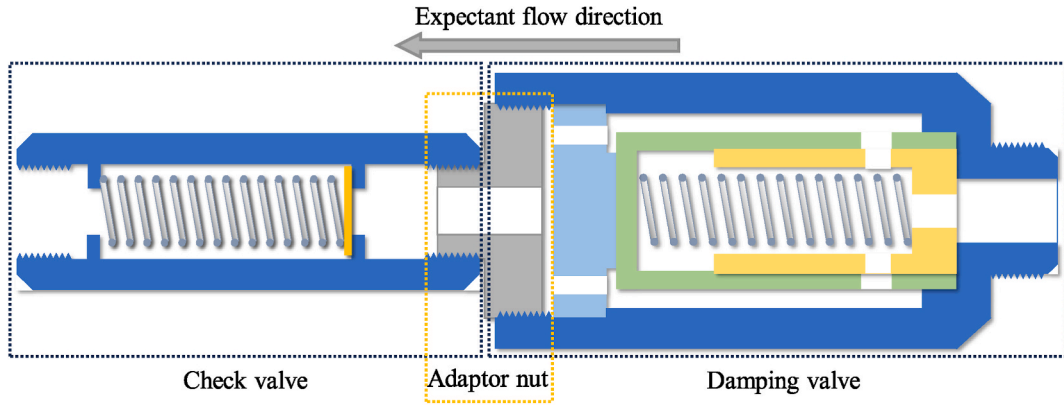


Fig. 4. Control valve.

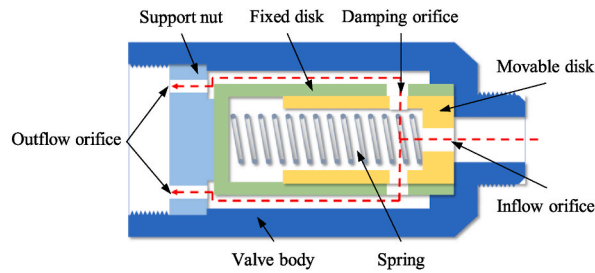


Fig. 5. Construction of the damping valve.

2.3. Working principle of VDVD

The working mechanism of the VDVD is shown in Fig. 8. When the piston rod moves to the left side Fig. 8 (a)), the oil in the left chamber flows into the external pipe and control valves in both sides of the cylinder. As the control valve was assembled in contrasting directions, the oil can only flow through the upper control valve to another chamber, while the oil in the lower control valve is obstructed by the check valve. Upon the piston rod traversal in the opposing direction (illustrated in Fig. 8 (b)), the oil flows from the lower control valve and is blocked by the upper control valve. In this configuration, two control valves in opposite directions ensure that VDVD can generate variable damping coefficients separately during varying directions of motion.

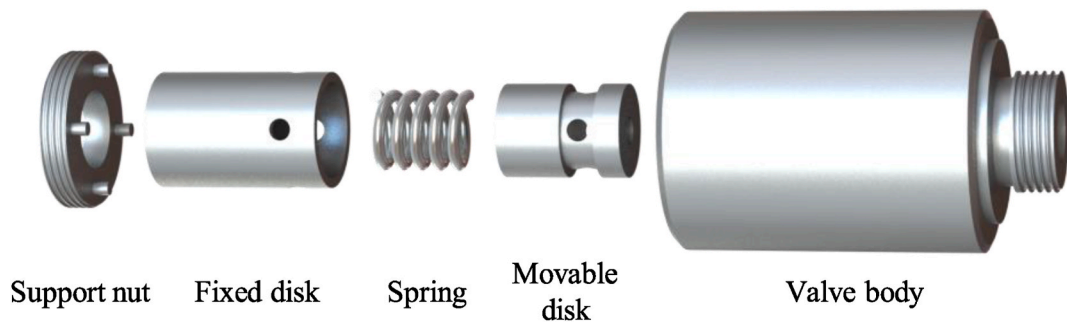


Fig. 6. The components of the damping valve.

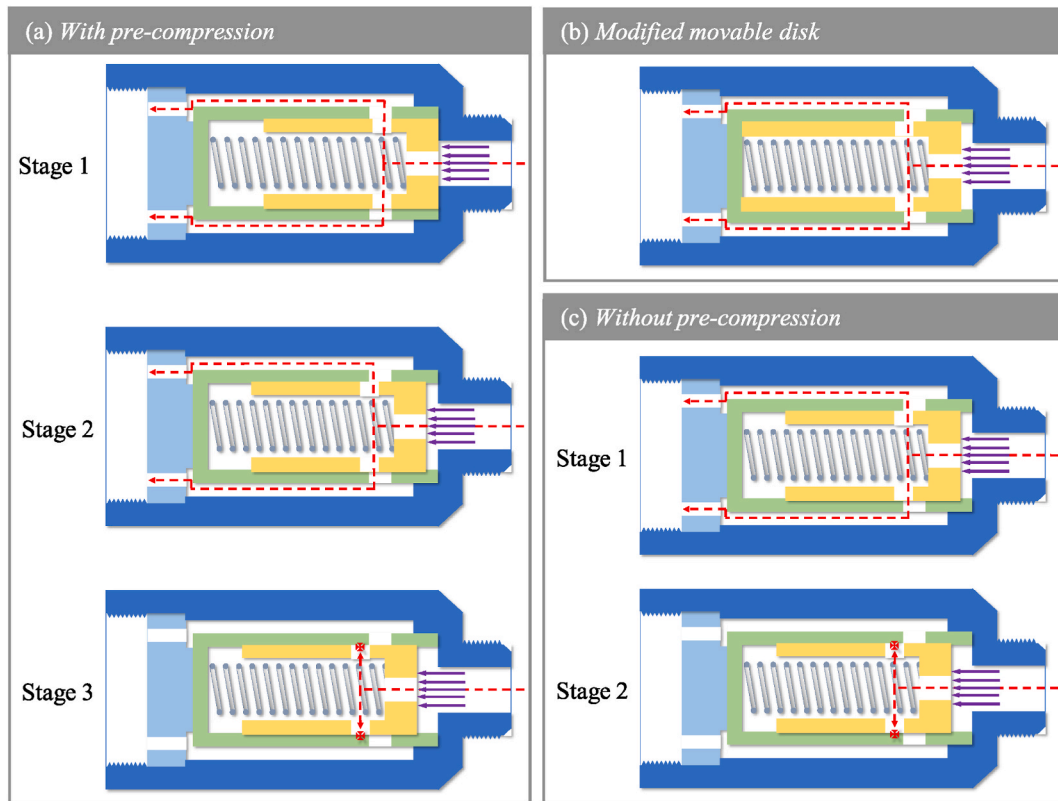


Fig. 7. Different working stages of damping valve: (a) spring with pre-compression; (b) modified movable disk; (c) spring without pre-compression.

3. Characteristic test of VDVD

The VDVD was fabricated and tested based on the design philosophy of it in this section. The test results are analysed and compared with CDVD in terms of hysteretic curves, maximum forces, and ability of energy dissipation.

3.1. Test scheme and set-up

The characterization tests of VDVD were conducted in the structural testing lab, Qingdao University of Technology, with the layout shown in Fig. 9. The actuator used in this test is a 50 tons dynamic hydraulic servo actuator with a load cell and displacement sensors, produced by MTS, USA. The displacement and force results were recorded by the load cell and displacement sensor in the actuator respectively. The actuator and the VDVD are connected by an adaptor plate and the assembly was placed in the reaction frame vertically.

To investigate the performance of the VDVD, the sinusoidal displacement loading with frequencies ranging from 0.02 Hz to 0.25 Hz and amplitudes including 40 mm, 50 mm, 60 mm and 70 mm is used in the test. Besides, the testing shall be stopped if the load approaches the force limit of the actuator's maximum capacity or any unconventional behavior (significant oscillations of damper,

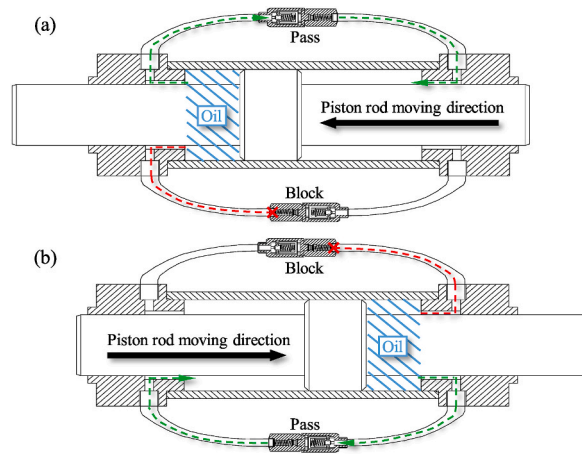


Fig. 8. Working principles of VDVD: (a) piston rod moves to the left direction; (b) piston rod moves to the right direction.

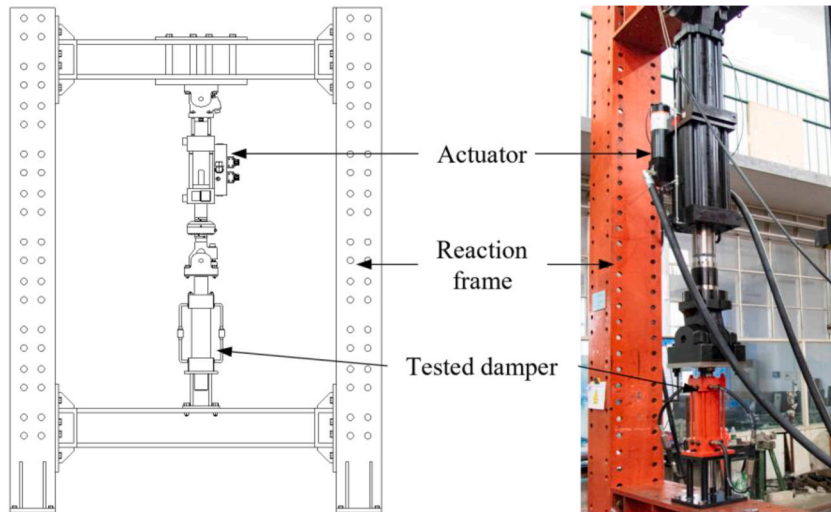


Fig. 9. Test set-up.

pronounced elevation in temperature of the damper and so forth) in the test. Furthermore, to investigate the influence of spring pre-compression, three springs with different wire diameters (i.e., stiffness and pre-pressure force) were chosen for the test. Table 1 shows the specifications of the three springs.

To directly compare the energy dissipation capacity with the CDVD, the movable disk of one damping valve was fixed, ensuring the area of the damping orifices in this valve will not be affected by the flow velocity. Upon the piston rod moving to a specific direction in which oil flows through the fixed control valve, the damping coefficient will remain constant and perform as a CDVD. Conversely, when the piston moves in another direction, the oil will flow through another control valve and perform the character of variable damping. As such, half of the force-displacement loop represents the characteristics of the VDVD, and the other half represents the CDVD.

Table 1
Basic information of springs.

Spring	Size (mm) (Wire diameter × Outside diameter × Length)	Stiffness (N/mm)	Compression length (mm)	Pre-pressure (N)
S-1.4	1.4 × 12 × 35	56.2	5	281.1
S-1.8	1.8 × 12 × 35	105.1	5	525.5
S-2.0	2.0 × 12 × 35	144.6	5	723.0

3.2. Test results

3.2.1. Hysteretic curves

At the beginning of the test, the VDVD was tested at a relatively low frequency (i.e., low velocity) of 0.02Hz to characterize the friction force of the damper. Fig. 10 presents the force-displacement loops of VDVD at 0.02Hz. For the hysteretic curves in this section, the solid lines are the test results, while the dot lines represent the numerical model results which are introduced in section 4.1 in detail. The friction force of the damper was set as 1.2 kN based on the average static friction force from Fig. 10. It is worth noting that all test results below in section 3.2 are all original data without subtracting friction force.

The force-displacement curves of the VDVD with S-1.4 spring are shown in Fig. 11 by the solid lines, which are under excitation frequencies between 0.03Hz and 0.11Hz, respectively. During the test, the external pipes wobble up and down at high loading velocities due to a lack of external restrictions. For safety purposes, we stopped the test and strengthened the external pipes and control values. So that the test results with frequencies higher than 0.11Hz are not obtained for this case.

It can be seen in Fig. 11 (a) ~ (e) that the force-displacement curves of the VDVD closely parallel the viscous damper with a constant damping coefficient. For those cases, the liquid pressure at the inflow orifice is less than the pre-pressure force of the spring. As the loading velocity increases, as shown in Fig. 11 (f) ~ (i), the damping force exhibits a significant increase in the lower half of the loops, whereas the upper half remains the shape of CDVD. According to the phenomenon of loops, it can be considered that in a constant velocity, the liquid pressure at the inflow orifice is higher than the pre-pressure force of spring, resulting in the decrease of the area of damping orifices and the increase of damping coefficient and damping force. Consequently, in contrast to the upper loop segments, the lower loop segments evince heightened energy dissipation capabilities.

Fig. 12 shows the hysteretic curves of VDVD with S-1.8 spring. The results with frequencies lower than 0.08Hz are not presented here as those results show the same characteristics as those between 0.08Hz and 0.12Hz, which are the same as CDVD. While the hysteretic curves at 0.19Hz (Fig. 12 (l)) for all amplitudes exhibit the variable damping characteristics. Therefore, the test was stopped after 0.19Hz to prevent the force from exceeding the limitation of the actuator when the damping orifices were totally closed.

The curves in Fig. 12 show a similar shape as that in Fig. 11. As the S-1.8 spring has a larger spring pre-compression force than that of S-1.4 spring, the critical velocity at which the damping coefficient starts to change increases, transitioning from 0.08 Hz, 70 mm loading (maximum velocity of loading: 35.17 mm/s) as depicted in Fig. 11 (f), to 0.13 Hz, 70 mm loading (maximum velocity of loading: 57.15 mm/s) as illustrated in Fig. 12 (f). As can be seen in Fig. 12 (l), the loading velocity is large enough that all the loops at different displacements exhibit the variable damping phenomenon at lower half segments which shows a significant force and energy dissipation difference compared with the upper half segments.

Fig. 13 gives the force-displacement curves of VDVD with S-2.0 spring with different loading frequencies and amplitudes, ranging from 0.14Hz to 0.25Hz. Results with frequencies lower than 0.14Hz are not presented here as those results are all looked at as same as CDVD. Fig. 13 shows that the damping force rises dramatically as the loading amplitudes and frequencies grow beyond 70 mm and 0.18Hz respectively in the lower half segments, while a slight increase can also be observed in the upper half segments, albeit negligible when contrasted with the lower counterparts.

However, the absolute peak force within the lower half segment of the hysteresis curve exhibits a gradual diminution at relatively high frequencies. As stated before, a larger damping force at high velocities is caused by the reduction in the area of damping orifices. For passive viscous damper, the majority of energy dissipation occurs through friction between the oil and the damping orifices, culminating in a temperature rise within the oil. This elevated temperature subsequently leads to a reduction in the kinematic viscosity of the oil, culminating in a decrease in damping force [29,30].

Since the excitation used in this test is the sinusoidal displacement wave, the maximum velocity should occur when the displacement is zero. However, from the above force and displacement loops, the max force of test results does not occur at the rest position (displacement equals zero) compared with the ideal results, as shown in Fig. 14. In other words, the maximum force does not occur at the maximum velocity. Note that the blue arrows indicate the loop's direction. Meanwhile, at the beginning of the damping force increase, the damping force shows an unusual discontinuity increase. Upon the inspection of the damping valve after the test, we found that during the deep compression of the spring, owing to the machining error of the valve, the movable disk did not move

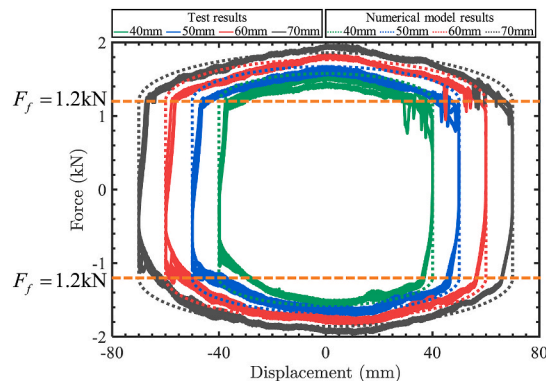


Fig. 10. Force-displacement curves of VDVD with S-1.4 spring at 0.02Hz.

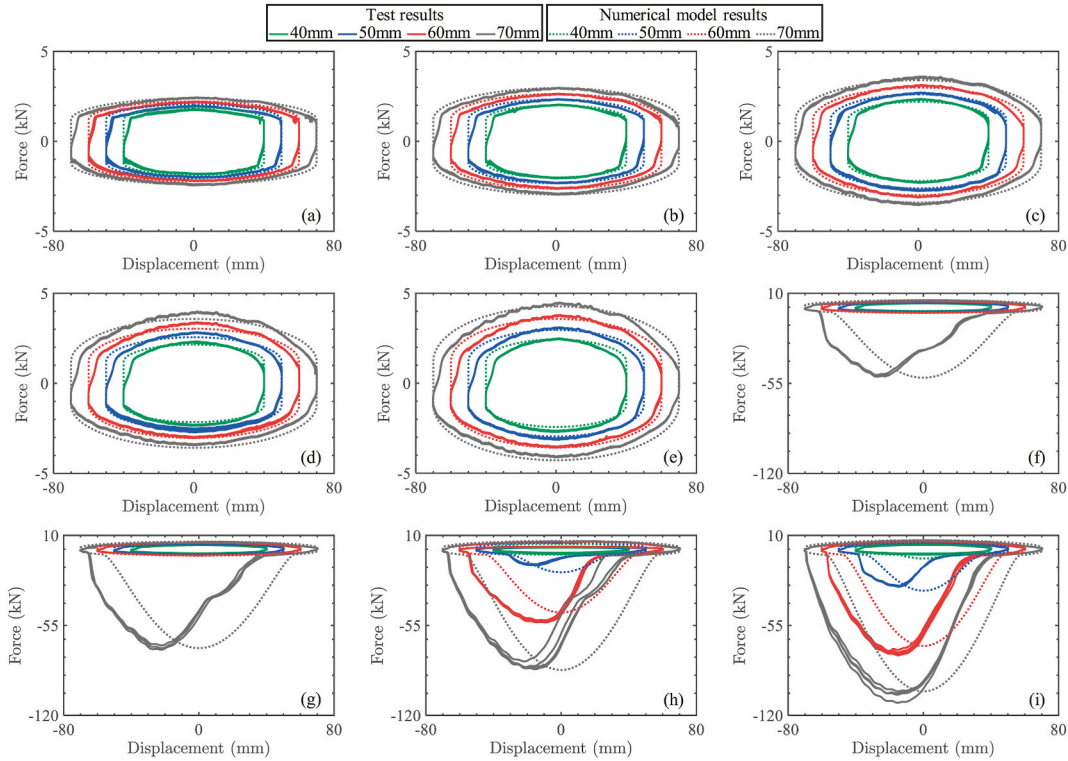


Fig. 11. Force-displacement curves of VDVD with S-1.4 spring: (a) 0.03 Hz; (b) 0.04 Hz; (c) 0.05 Hz; (d) 0.06 Hz; (e) 0.07 Hz; (f) 0.08 Hz; (g) 0.09 Hz; (h) 0.10 Hz; (i) 0.11 Hz.

smoothly, which led to damping orifices maintaining a fixed area. This occurrence contributes to a discontinuity increase of damping force and an asymmetry in force distribution. This phenomenon could be solved by improving the manufacturing process.

3.2.2. Maximum force

As the maximum force in different velocities of a damper is a vital index to evaluate the performance of the damper, it is important to find out the differences between variable damping damper and constant damping damper. Fig. 15 shows the correlation between the peak damping force and the velocity. Notably, the peak damping forces are derived from hysteretic curves in conjunction with their corresponding velocities (obtained through displacement derivation). The peak variable damping forces and peak constant damping force are obtained from the absolute value of the lower half segment and upper half segment of hysteretic respectively.

As shown in Fig. 15, when the loading velocity remains small (<31.2 mm/s in Fig. 15 (a), <56.5 mm/s in Fig. 15 (b), <79.1 mm/s in Fig. 15 (c)), the slope of curves, i.e., the damping coefficient, remains constant. This is irrespective of the spring stiffness and the pre-pressure force applied to the spring. At this moment, the oil pressure at the inflow orifice does not reach the spring pre-pressure, and the area of the damping orifices retains its original dimensions. Consequently, no differences between the VDVD and CDVD are found.

However, as the velocity increases, the damping force exhibits a non-linear and substantial increase once both velocity and force reach thresholds, referred to as the critical velocity and critical force, respectively. At the maximum velocity shown in each figure in Fig. 15, the ratio of variable damping force to constant damping force is 17.0, 9.8 and 8.7 respectively. Thus, the VDVD can provide a substantially greater damping force than the CDVD, even if the initial design parameters of the two kinds of damper are the same.

Fig. 16 delineates the comparison between the critical values (force and velocity) of the VDVD employing various springs in relation to those with the S-1.4 spring, as well as the ratio of pre-pressure force applied to the spring. This figure illuminates that the ratio of critical force and velocity closely parallels the ratio of spring pre-pressure force. Consequently, in the absence of pre-pressure within the spring, the VDVD will exhibit its inherent damping coefficient characteristics upon the application of an external loading.

3.2.3. Energy dissipation

The capacity for energy dissipation stands as another pivotal parameter in assessing a damper's performance. As the special setup to compare the constant damping and variable damping intuitively, the values of energy dissipation for 'ith' cycle of constant damping segment $E_{c(i)}$ and variable damping segment $E_{v(i)}$ are computed by Equations (1) and (2) respectively and illustrated in Fig. 17, where the subscript iff $F \geq 0$ and iff $F < 0$ means if and only if F is greater or equal to zero and less than zero respectively. The cumulative dissipated energy is the cumulative energy value of the first cycle of each loading from the loading protocol, calculated by equations (3) and (4) for cumulative energy of constant damping $E_{cc(i)}$ and variable damping $E_{cv(i)}$ respectively. The cumulative dissipated energy of VDVD and CDVD are presented in Fig. 18 respectively.

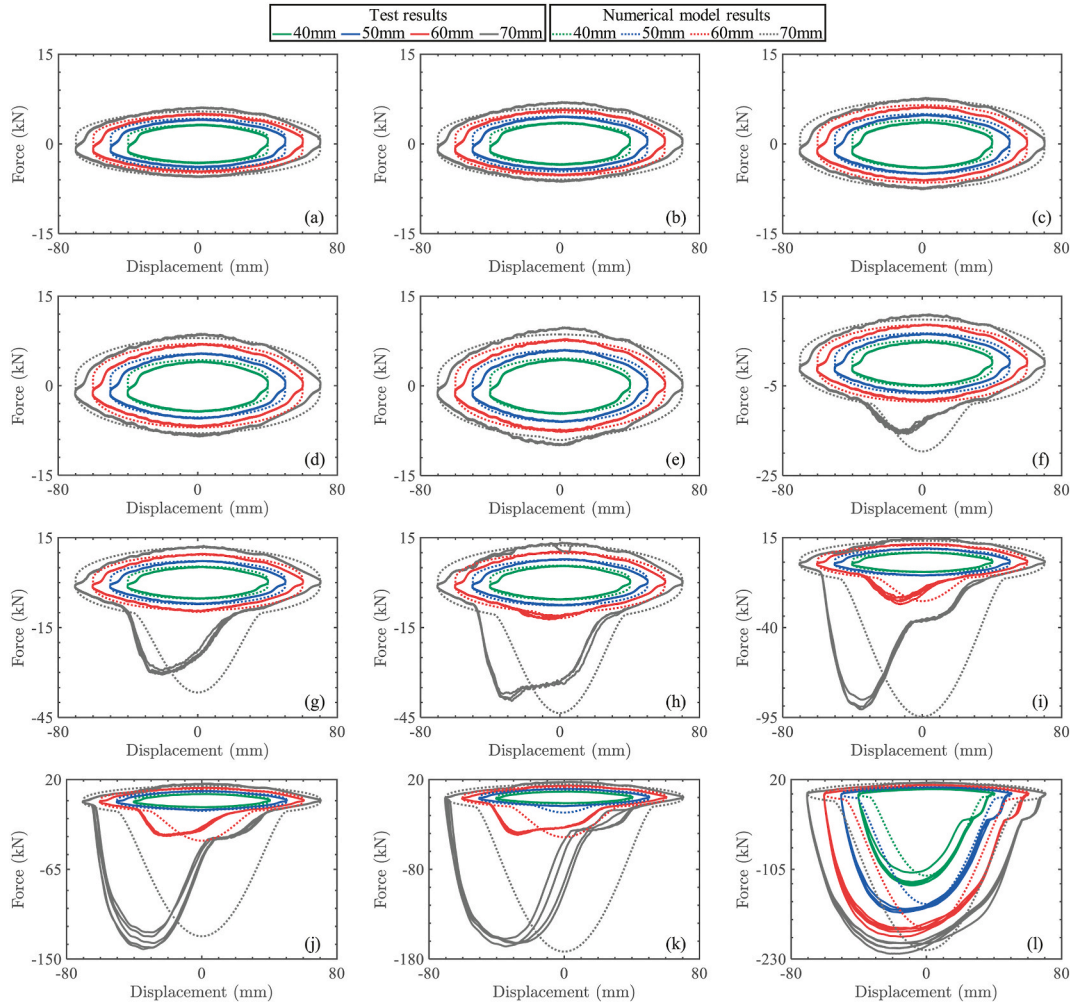


Fig. 12. Force-displacement curves of VDVD with S-1.8 spring: (a) 0.08 Hz; (b) 0.09 Hz; (c) 0.10 Hz; (d) 0.11 Hz; (e) 0.12 Hz; (f) 0.13 Hz; (g) 0.14 Hz; (h) 0.15 Hz; (i) 0.16 Hz; (j) 0.17 Hz; (k) 0.18 Hz; (l) 0.19 Hz.

$$E_{c(i)} = \int F_{iff} F_{\geq 0} dx \quad (1)$$

$$E_{v(i)} = \int |F_{iff} F_{< 0}| dx \quad (2)$$

$$E_{cc(i)} = \sum_{k=1}^i E_{c(k)}, \text{ for } i = 1, 2, 3, \dots, N \quad (3)$$

$$E_{cv(i)} = \sum_{k=1}^i E_{v(k)}, \text{ for } i = 1, 2, 3, \dots, N \quad (4)$$

As evident from Fig. 18, the energy dissipation capability of the VDVD significantly surpasses that of the CDVD, displaying energy ratios of 4.8, 3.1, and 2.3, respectively, after the final cycle. This reaffirms the superior energy absorption capacity of the VDVD when compared to the constant damping damper.

4. Methodology of wind vibration control analysis

The superb efficacy of viscous dampers in mitigating vibrations induced by the wind has been substantiated. To assess the effectiveness of the VDVD in wind-induced vibration control, a numerical model of the VDVD is proposed and verified in this section. The analysis method and building model as well as wind loadings are also introduced.

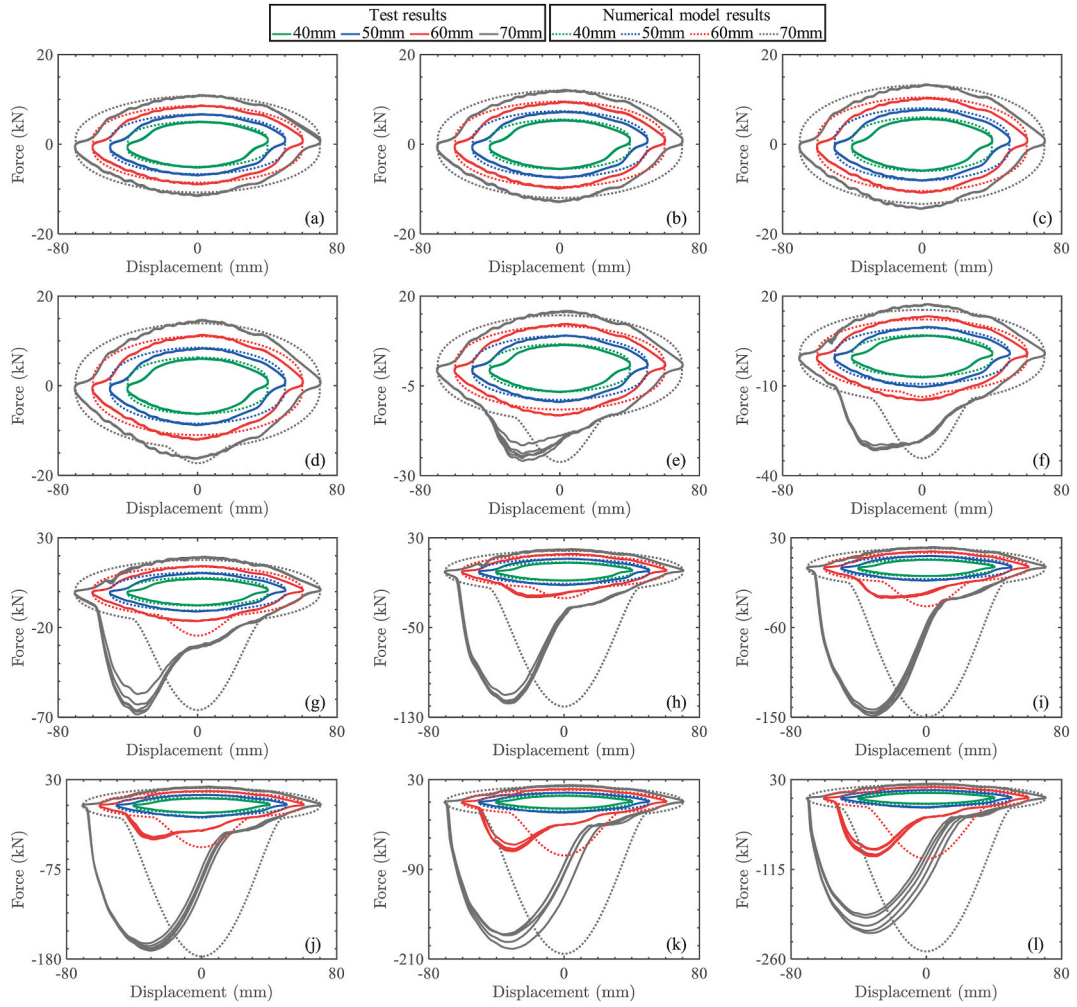


Fig. 13. Force-displacement curves of VDVD with S-2.0 spring: (a) 0.14 Hz; (b) 0.15 Hz; (c) 0.16 Hz; (d) 0.17 Hz; (e) 0.18 Hz; (f) 0.19 Hz; (g) 0.20 Hz; (h) 0.21 Hz; (i) 0.22 Hz; (j) 0.23 Hz; (k) 0.24 Hz; (l) 0.25 Hz.

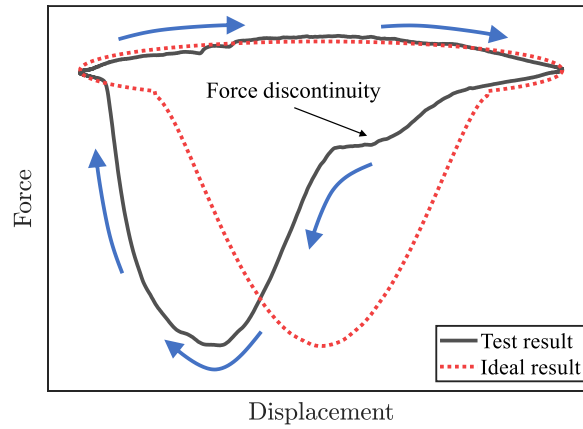


Fig. 14. Loading direction and comparison of the test result and the ideal result.

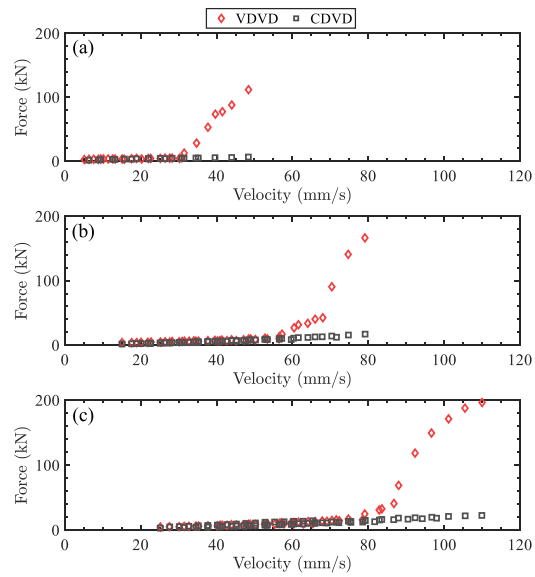


Fig. 15. Relationship between peak force and velocity: (a) S-1.4 spring; (b) S-1.8 spring; (c) S-2.0 spring.

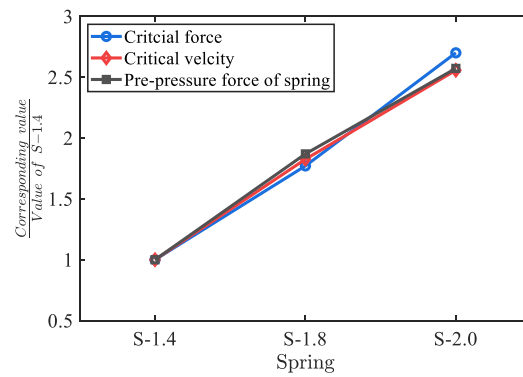


Fig. 16. Comparison of the ratio of critical value and pre-pressure force of spring.

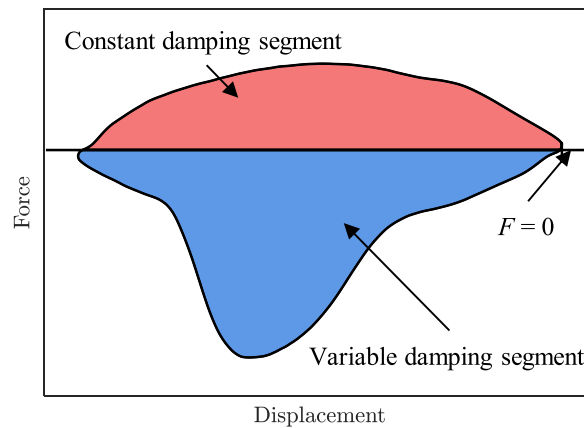


Fig. 17. Definition of different segments of hysteretic loops.

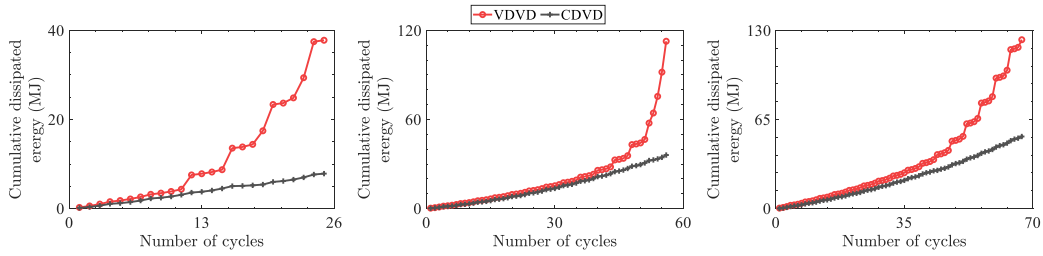


Fig. 18. Cumulative dissipated energy of variable damping and constant damping: (a) S-1.4 spring; (b) S-1.8 spring; (c) S-2.0 spring.

4.1. Numerical model of VDVD and CDVD

The test results show that the damping coefficient of VDVD exhibits variability in response to loading velocity, showcasing superior dissipative capabilities, particularly at high velocities. In order to investigate the performance of variable damping and simplify the analysis, the power function is employed to model the damping coefficient-velocity relationship. The damping coefficient model of VDVD (C_v) and CDVD (C_c) are given in equation (5) and equation (7) respectively. Equations (6) and (8) present the corresponding damping force model of VDVD (F_v) and CDVD (F_c) respectively.

$$C_v = \begin{cases} c_1 \cdot (|v| - v_c)^p + c_0 & |v| > v_c \\ c_0 & |v| \leq v_c \end{cases} \quad (5)$$

$$F_v = \begin{cases} \{c_0 \cdot v_c + [c_1 (|v| - v_c)^p + c_0] \cdot (|v| - v_c)^m + F_f\} \cdot \text{sgn}(v) & |v| > v_c \\ c_0 \cdot v^m + F_f \text{sgn}(v) & |v| \leq v_c \end{cases} \quad (6)$$

$$C_c = c_0 \quad (7)$$

$$F_c = C_c \cdot v = c_0 \cdot v^m + F_f \text{sgn}(v) \quad (8)$$

Where c_0 is the constant damping coefficient which is related to the initial design parameters of the damper such as the area of damping orifices, the inside dimension of the cylinder and external pipes and so forth; c_1 is the variable damping parameter, p is the exponent of power function, both of the two parameters are related with the design of control valve that control the rate of variable damping; v is the velocity between the cylinder and piston rod of damper; v_c represents the critical velocity that damping coefficient start changing which related to the pre-pressure force of spring; m denotes the viscosity of the fluid, encompassing shear-thickening fluid, shear-thinning fluid, and Newtonian fluid, corresponding to values greater than, less than, and equal to 1, respectively.

In equation (5), the damping coefficient of VDVD is divided into two distinct segments. When the absolute value of velocity falls below the critical velocity, the damping coefficient maintains a constant value equivalent to c_0 . Conversely, as the absolute value of velocity is greater than the critical velocity, the damping coefficient increases nonlinearly commencing from c_0 . The constant damping coefficient of c_0 is used in CDVD to balance the initial performance of two dampers. Moreover, it is assumed that the oil in both dampers adheres to the characteristics of a Newtonian fluid, hence m equals 1. The damping orifices are assumed to progressively diminish with the input velocity and would not close completely, thereby the damping coefficient will increase continuously.

Given the intricate nature of the design parameters governing the VDVD, formulating a theoretical calculation method for parameters c_0 , c_1 , p and v_c presents a formidable challenge. To surmount this challenge, the curve fitting toolbox in MATLAB is employed to derive these parameters based on the peak force-velocity relationship shown in Fig. 15. The fitting parameters are listed in Table 2. The comparisons of test results and the numerical model results are depicted in the form of force-displacement and absolute maximum force-velocity relationships in Figs. 11–13 and 19 respectively.

As illustrated in Figs. 11–13 and 19, the outcomes generated by the proposed numerical model exhibit commendable concordance with the test results, except the time-lag effects caused by the reason discussed in Fig. 14. This alignment engenders confidence in the numerical model's reliability and, consequently, establishes a sound foundation for further analyses in the wind-induced vibration control assessment.

4.2. Building model and wind loading

The building model is a benchmark building with a 306m height, 76-story frame-core wall structure [31]. This benchmark model was proposed in the Second International Vibration Control Conference, and it is mainly used to study the effectiveness and

Table 2
Fitting parameters of VDVD numerical models.

Case	Critical velocity v_c (mm/s)	Constant damping coefficient c_0 (kN-s/mm)	Variable damping parameter c_1	Exponent p	Friction force F_f (kN)
S-1.4	26.3	0.1532	0.5497	0.7036	1.2
S-1.8	52.1				
S-2.0	73.8				

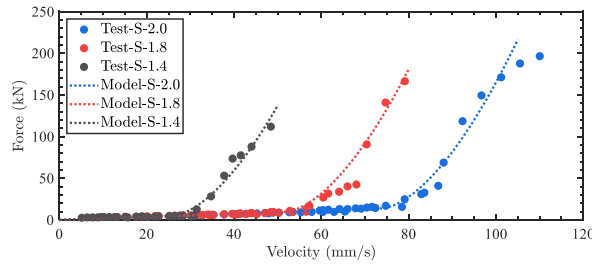


Fig. 19. Comparison of test results and numerical model results by absolute maximum force-velocity relationship.

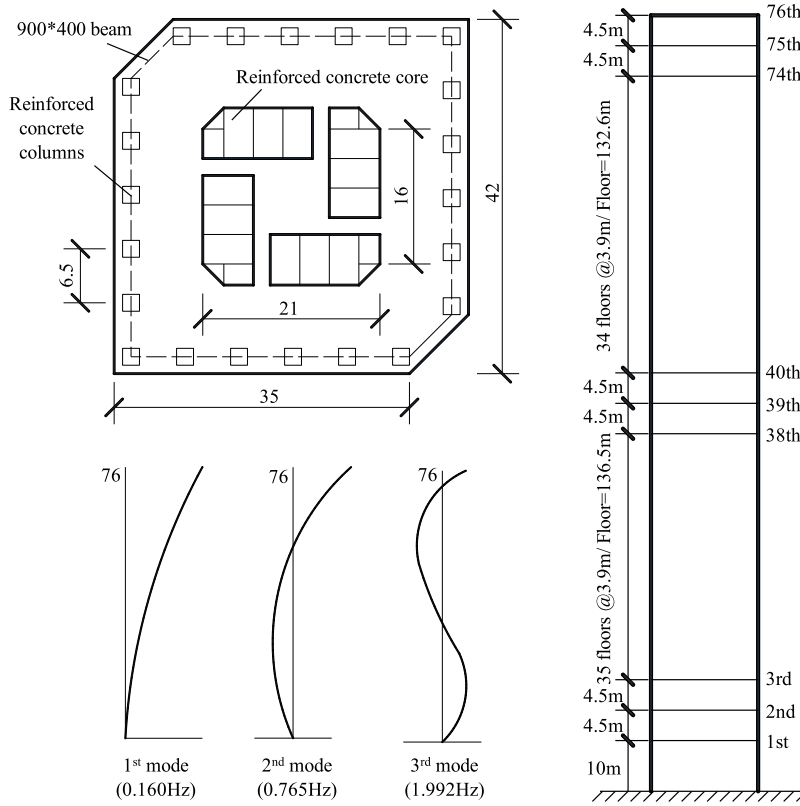


Fig. 20. The 76-story benchmark model and first three mode shapes [31].

applicability of different control methods and strategies under wind-induced vibration [32–34]. The information on the benchmark model and the mode shapes of the first three modes is shown in Fig. 20.

The wind loading used in the analysis is the wind tunnel test results of the 76-story rigid structure model conducted by Samali et al. in the wind tunnel laboratory of the University of Sydney [35]. Based on the test results, the wind loading time history of the original structure for 1 h can be obtained, and the wind loading under different wind speeds can be obtained by $(v_z/47.25)^2$. In the Load Code for the Design of Building Structure of China [36], the relationship between the wind pressure and the wind speed at 10m is shown in equation (9).

$$\omega_0 = \frac{1}{2} \rho v_0^2 \quad (9)$$

The relationship of wind speed at different heights is presented in equation (10).

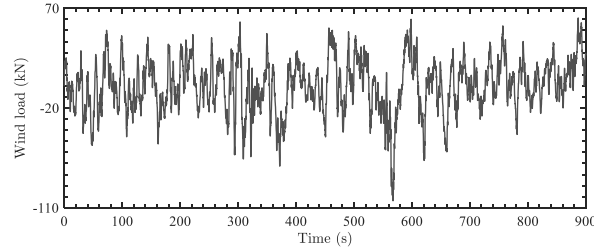
$$v_0 = v_z \left(\frac{10}{z} \right)^a \quad (10)$$

Where z is the altitude, v_0 is the wind speed at 10m; v_z is the wind speed at z ; ω_0 is the wind pressure; a is the surface roughness; and ρ is

Table 3

Basic information on wind excitations.

Case	Wind pressure (kN/m ²)	Wind speed at 10m (m/s)	Wind speed at 300m (m/s)	Multiply factor
1	0.3	21.91	46.30	0.96
2	0.6	30.98	65.47	1.92
3	0.9	37.95	80.20	2.88

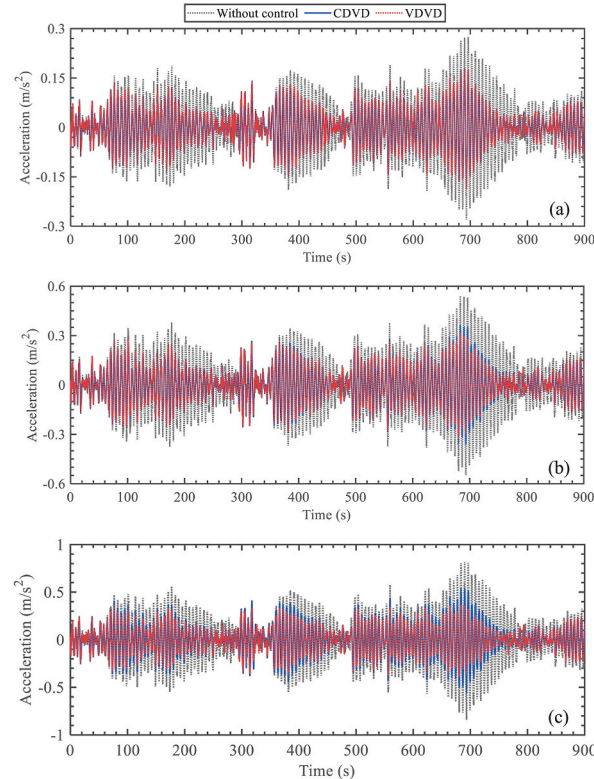
**Fig. 21.** Time history curve of wind loading at wind pressure of 0.3 kN/m² on the 76th DOF.

air density. In this section, the air density ρ and surface roughness are chosen as $1.25 \times 10^{-3} \text{ t/m}^3$ and 0.22 respectively.

The information of three wind excitations with different wind pressures is listed in Table 3. Fig. 21 illustrates the wind loading time-history curve at a wind pressure of 0.3 kN/m² on the 76th DOF.

4.3. Response analysis method

The 76-story building model is simplified as a 76-DOF linear model, in which structural matrices can be found in Ref. [31]. The state-space function is employed to generate the equation of motion of the system, which is solved by the ode15s solver in a variable time increment step in MATLAB. The equation of motion of the system is given by equation (11).

**Fig. 22.** Absolute (to the ground) acceleration time-history curves of the top floor in different wind pressures: (a) wind pressure at 0.3 kN/m²; (b) wind pressure at 0.6 kN/m²; (c) wind pressure at 0.9 kN/m².

$$[M]\{\ddot{X}(t)\} + [C]\{\dot{X}(t)\} + [K]\{X(t)\} = \{H\}\{F_w(t)\} - \{\Gamma\}\{F_d(t)\} \quad (11)$$

where $[M]$, $[K]$ and $[C]$ are the mass, stiffness and damping matrix respectively; $\{\ddot{X}(t)\}$, $\{\dot{X}(t)\}$ and $\{X(t)\}$ are the acceleration, velocity and displacement vector at time t . $\{F_w(t)\}$ and $\{F_d(t)\}$ is the wind loading vector and damping force vector at each DOF at time t . F_d can be obtained from equations (6) and (8). $\{H\}$ and $\{\Gamma\}$ is the position vector of wind loading and damper. As the ideal placement of the damper is not the central concern in this section, dampers are assigned to each layer of the building. The excitation velocity of the damper equals the interlayer velocity at the specific location where the damper is situated.

The state-space function form of equation (11) can be expressed as

$$\dot{z} = Az + BF \quad (12)$$

where $z = [X(t) \ \dot{X}(t)]^T$, $A = \begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}C \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ -M^{-1}\Gamma & M^{-1}H \end{bmatrix}$, $F = \begin{bmatrix} F_d(t) \\ F_w(t) \end{bmatrix}$, I and 0 are the identity and zero matrices of convenient size.

5. Wind vibration control performance evaluation

In this section, the performance of the VDVD is compared with that of the conventional viscous damper using the methodology in section 4, thereby accentuating its exceptional prowess.

As the fitting parameter of the numerical model of VDVD and CDVD in Table 2 is not tailored to the building model, the damping coefficient of CDVD is set as 2000 N s/m, the constant damping parameter, variable damping parameter and exponent of VDVD are chosen as 2000 N s/m, and 5×10^6 and 0.5 respectively after several trial calculations, which philosophy is also adopted by Ref. [10]. The friction force is taken as zero as it is neglectable compared with the damping force.

5.1. VDVD with the constant damping coefficient stage

In this part, the critical velocity v_c for all the VDVDs is chosen as 4.3 mm/s which is larger than the maximum interlayer velocity subjected to the wind loading at wind pressure of 0.3 kN/m². Figs. 22 and 23 depict the absolute (to the ground) acceleration and displacement time-history curves of the top floor in different wind excitations. Tables 4 and 5 provide the absolute peak value of

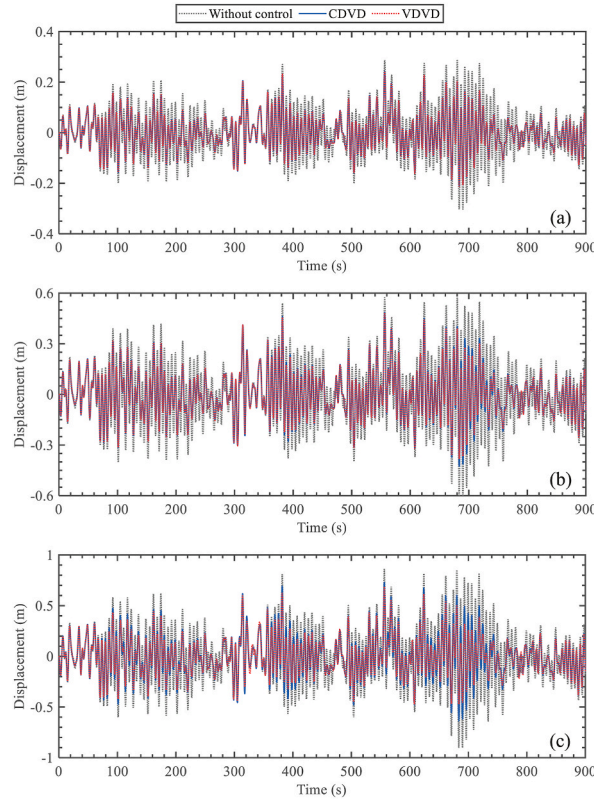


Fig. 23. Absolute (to the ground) displacement time-history curves of the top floor in different wind pressures: (a) wind pressure at 0.3 kN/m²; (b) wind pressure at 0.6 kN/m²; (c) wind pressure at 0.9 kN/m².

Table 4

Absolute peak value of acceleration responses and response reductions of the building.

Floor	Without control			VDVD						CDVD			
	0.3 kN/m ²	0.6 kN/m ²	0.9 kN/m ²	0.3 kN/m ²		0.6 kN/m ²		0.9 kN/m ²		0.3 kN/m ²	0.6 kN/m ²	0.9 kN/m ²	Acc. Red. (%)
	Acc. (mm/s ²)	Acc. (mm/s ²)	Acc. (mm/s ²)	Acc. (mm/s ²)	Red. (%)	Acc. (mm/s ²)	Red. (%)	Acc. (mm/s ²)	Red. (%)	Acc. (mm/s ²)	Acc. (mm/s ²)	Acc. (mm/s ²)	
30	63.37	126.75	190.12	40.66	35.85	75.61	40.35	93.12	51.02	40.66	81.31	121.97	35.85
40	97.49	194.97	292.46	60.44	38.00	112.70	42.20	141.19	51.72	60.44	120.88	181.31	38.00
50	141.63	283.26	424.88	87.13	38.48	166.77	41.12	205.82	51.56	87.13	174.26	261.39	38.48
60	184.32	368.64	552.96	116.58	36.75	221.35	39.95	275.17	50.24	116.58	233.15	349.73	36.75
70	236.82	473.64	710.46	154.66	34.69	280.16	40.85	336.93	52.58	154.66	309.33	463.99	34.69
76	279.53	559.06	838.58	186.04	33.45	326.42	41.61	386.39	53.92	186.04	372.07	558.11	33.45

Table 5

Absolute peak value of displacement responses and response reductions of the building.

Floor	Without control			VDVD						CDVD			
	0.3 kN/m ²	0.6 kN/m ²	0.9 kN/m ²	0.3 kN/m ²		0.6 kN/m ²		0.9 kN/m ²		0.3 kN/m ²	0.6 kN/m ²	0.9 kN/m ²	Disp. Red. (%)
	Disp. (mm)	Disp. (mm)	Disp. (mm)	Disp. (mm)	Red. (%)	Disp. (mm)	Red. (%)	Disp. (mm)	Red. (%)	Disp. (mm)	Disp. (mm)	Disp. (mm)	
30	64.13	128.27	192.40	53.54	16.51	106.84	16.70	152.61	20.68	53.54	107.08	160.63	16.51
40	106.59	213.17	319.76	88.29	17.16	176.17	17.36	251.46	21.36	88.29	176.59	264.88	17.16
50	155.51	311.01	466.52	127.62	17.93	254.61	18.14	363.11	22.17	127.62	255.24	382.85	17.93
60	209.42	418.84	628.26	170.25	18.70	339.64	18.91	483.97	22.97	170.25	340.50	510.75	18.70
70	266.29	532.58	798.86	214.56	19.43	427.99	19.64	609.39	23.72	214.56	429.12	643.68	19.43
76	302.63	605.26	907.89	242.75	19.79	484.19	20.00	689.14	24.09	242.75	485.50	728.26	19.79

acceleration and displacement responses and corresponding response reductions of the building under varying wind loadings respectively. One thing that should be noted is that the response reductions for CDVD under different wind loading intensities are identical as the results are obtained based on linear analysis. Therefore, the response reductions of CDVD under different loading intensities are only presented in the last column of [Tables 4 and 5](#).

As shown in the above chart, based on the absolute peak structural responses and response reductions in 0.3 kN/m² wind loading, the control efficacy of VDVD aligns with that of CDVD when the critical velocity is sufficiently high to maintain VDVD in a state of constant damping coefficient. At this stage, the variable damping properties of VDVD do not participate in controlling the building, thereby enabling its design to mirror that of CDVD and furnish equivalent control performance.

As the wind pressure increases, the displacement and acceleration absolute peak response reductions of VDVD exhibit a continuous increase trend. The reduction in absolute peak acceleration of the top floor increases from 33.45 % in 0.3 kN/m² wind loading to 53.92 % in 0.9 kN/m² wind loading, while the reduction in absolute peak displacement increases from 19.79 % to 24.09 %. [Fig. 24](#) illustrates the damping force-interlayer displacement curves, damping force-interlayer velocity curves, and cumulative dissipated energy of the VDVD and CDVD of the top floor under wind loadings of three wind pressures. It is evident in [Fig. 24 \(a\)](#) that the hysteretic curves of VDVD mirror those of CDVD at 0.3 kN/m² wind loading. At 0.3 kN/m² wind pressure, the damping coefficient (slope of curves in [Fig. 24 \(b\)](#)) remains constant as CDVD, and the cumulative dissipated energy ([Fig. 24 \(c\)](#)) has no difference between the two dampers. When the building is subject to a stronger wind loading, VDVD shows the characteristics of variable damping after the critical velocity,

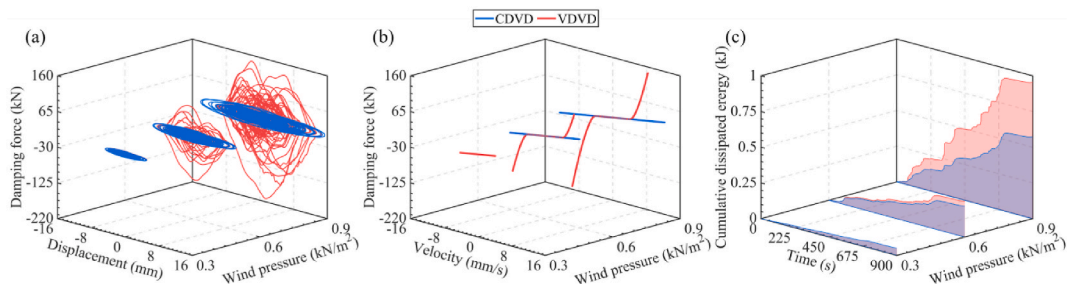


Fig. 24. Hysteretic curves and cumulative dissipated energy of CDVD and VDVD on the top floor: (a) damping force-interlayer displacement curves; (b) damping force-interlayer velocity curves; (c) cumulative dissipated energy.

resulting in a larger damping force and plumper hysteretic curves and higher energy dissipation. The cumulative dissipated energy ratios between VDVD and CDVD range from 1.0 to 1.74. Therefore, under identical initial designs, VDVD exhibits an equivalent performance with CDVD under minor wind loadings while offering a greater damping force and enhanced energy dissipation under strong wind loadings.

As the critical velocity will affect the engagement of variable damping, it is necessary to investigate the influence of critical velocity on the control performance of VDVD. The reductions in the absolute peak value of acceleration and displacement of the top floor under wind loadings with different wind pressures and critical velocities are depicted in Fig. 25.

From Fig. 25, it is evident that the response reduction of absolute peak values decreases with the critical velocity rises, while it exhibits an increment as the wind pressure shows upward trends at low critical velocities. In case when the critical velocity is large enough that the constant damping stage covers the loading process, the control performance is not affected by wind pressure (like the displacement response reduction when critical velocity equals 6 mm/s). On the contrary, in cases when the critical velocity is relatively small, the contribution of the constant damping stage is negligible. Although the damping coefficient of VDVD will increase continuously as the wind pressure rises (i.e., input velocity of damper), the response reduction only experiences a slight increase (like the response reduction when critical velocity equals 0 mm/s). This phenomenon arises from the fact that the damping ratio provided by VDVD for the entire building has already attained a substantial level, rendering any additional increase in damping inconsequential to the structure. However, for other situations, better control performance can be observed in high wind pressure as the damping coefficient is relatively high compared to situations of low wind pressure.

5.2. VDVD without the constant damping coefficient stage

Without the constant damping stage (i.e., critical velocity equals zero), the VDVD will exhibit the characteristic of variable damping in all velocity ranges. The time-history curves of absolute (to the ground) acceleration and displacement of the top floor under three wind pressures are given in Figs. 26 and 27 respectively. The absolute peak value of acceleration reduction rises from 64.6 % in wind pressure of 0.3 kN/m² to 68.8 % in wind pressure of 0.9 kN/m². While the control performance of displacement is also remarkable, increases from 38.5 % in wind pressure of 0.3 kN/m² to 43.2 % in wind pressure of 0.9 kN/m².

Fig. 28 gives the hysteretic curves and cumulative dissipated energy of CDVD and VDVD. It is evident that plumper hysteretic curves (Fig. 28 (a)) and higher energy dissipation (Fig. 28 (c)) of VDVD can be seen in wind pressure of 0.3 kN/m² as the damping coefficient (Fig. 28 (b)) increases dramatically from zero velocity. The cumulative dissipated energy ratios between VDVD and CDVD range from 2.7 to 3.14. By contrast, the hysteretic curves and cumulative dissipated energy of VDVD shown in Fig. 24 have no distinct differences with CDVD. It indicates the wide adjustment range of VDVD in force compared with CDVD.

To sum up, under identical initial designs, VDVD without a constant damping stage can offer a greater damping force and a superior control performance under wind loadings with any intensities in contrast with CDVD.

6. Conclusion

In this study, a novel variable damping viscous damper (VDVD) was proposed and tested. The VDVD was achieved through an innovative valve design in which a compressible spring works in conjunction with a movable disk to adjust the size of the damping orifice subjected to various loading velocities without external energy supplies. The tests were designed to compare the performance of constant damping viscous damper (CDVD) and VDVD in one test by artificially cancelling the variable damping function of one valve. The numerical models of VDVD were formulated and verified including springs with and without pre-pressure force. These proposed models were also employed in the analysis of wind-induced vibration control performance for a high-rise structure, drawing a sharp contrast between VDVD and CDVD. Some conclusions of this study can be given as follows.

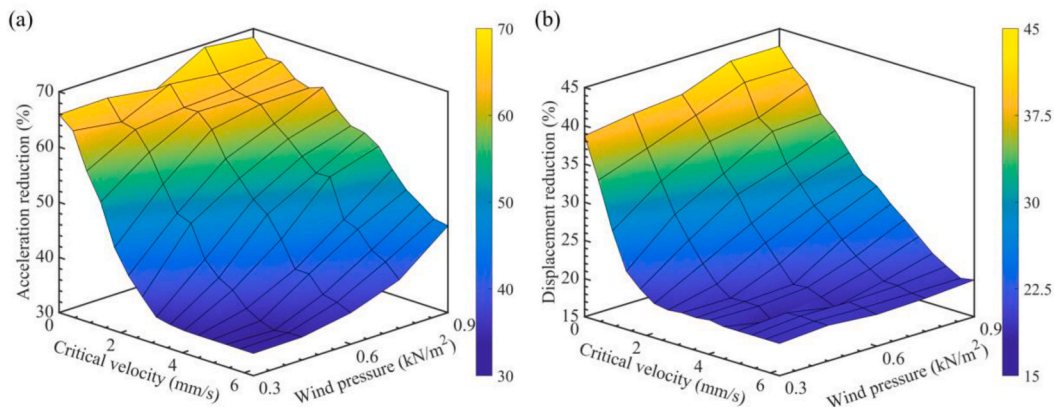


Fig. 25. Reduction in absolute peak value of response of the top floor: (a) acceleration response reduction; (b) displacement response reduction.

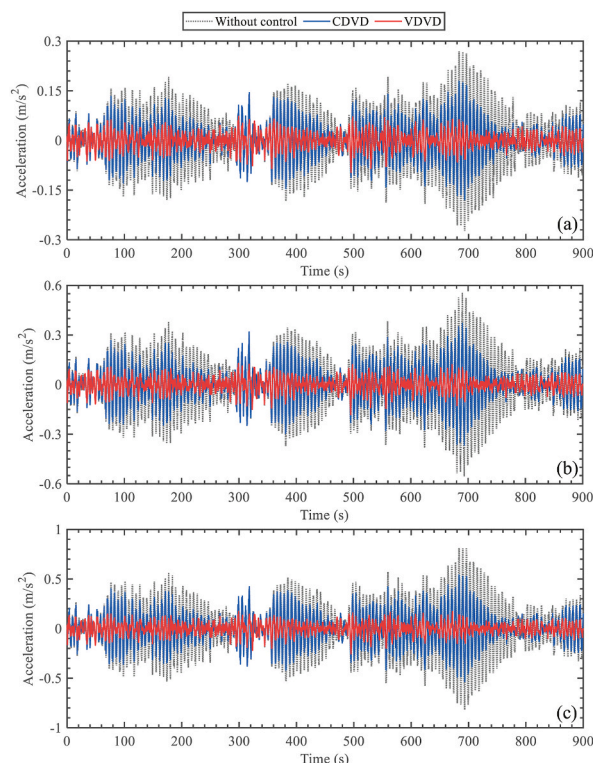


Fig. 26. Absolute (to the ground) acceleration time-history curves of the top floor in different wind pressures when VDVD without a constant damping stage: (a) wind pressure at 0.3 kN/m^2 ; (b) wind pressure at 0.6 kN/m^2 ; (c) wind pressure at 0.9 kN/m^2 .

- (1) The variable damping property was verified by the tests. With the innovative damping valve, the damping coefficient could be changed continuously by adjusting the area of damping orifices. In comparison with CDVD, VDVD exhibits an escalating damping coefficient with velocity, culminating in a more substantial damping force and an enhanced capacity for energy dissipation.
- (2) Through adjusting the stiffness and pre-pressure force of the spring, the critical force and velocity which distinguish the constant and variable damping coefficient could be fine-tuned. When a spring with pre-pressure force is integrated into the control valve, the VDVD exhibits the ability to achieve both constant and variable damping coefficients. As the loading velocity rises, the VDVD mirrors the performance of CDVD when the input velocity remains below the critical velocity. However, beyond this critical velocity, VDVD demonstrates the distinctive traits of variable damping. Conversely, when no pre-pressure force of spring was applied, the VDVD displayed its variable damping properties at any loading velocities.
- (3) Based on the test results, numerical models of VDVD which were calibrated using the power function were proposed. The numerical models account for initial damping coefficients, critical velocities, and variability of variable damping coefficients. The comparison of test results and numerical model results demonstrate good agreements, which were capable of modelling the force-velocity relationships of VDVD in the wind-induced vibration control analysis.
- (4) The evaluations of the wind vibration control performance of VDVD were conducted and compared with CDVD in a 76-story high-rise building. Two variants of VDVD were considered in this analysis (i.e., VDVD with and without a constant damping stage). The results exhibited that, under identical initial designs, the VDVD with a constant damping stage could perform an equivalent level of control efficacy as CDVD by an appropriate critical velocity design under minor wind loadings. In contrast, it provided a greater damping force and enhanced energy dissipation under strong wind loadings. For other cases that VDVD without a constant damping stage, it demonstrated a larger damping force and superior control performance against wind loadings of varying intensities, when compared with CDVD.

CRediT authorship contribution statement

Mao Li: Writing – original draft, Visualization, Software, Formal analysis. **Mingmei Shi:** Writing – review & editing, Visualization, Validation. **Weiqing Fu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

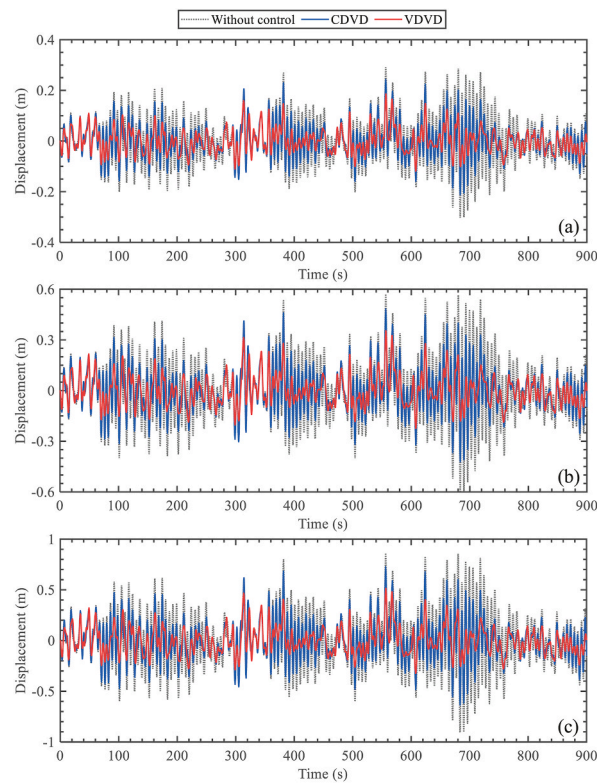


Fig. 27. Absolute (to the ground) displacement time-history curves of the top floor in different wind pressures when VDVD without a constant damping stage: (a) Wind pressure at 0.3 kN/m^2 ; (b) Wind pressure at 0.6 kN/m^2 ; (c) Wind pressure at 0.9 kN/m^2 .

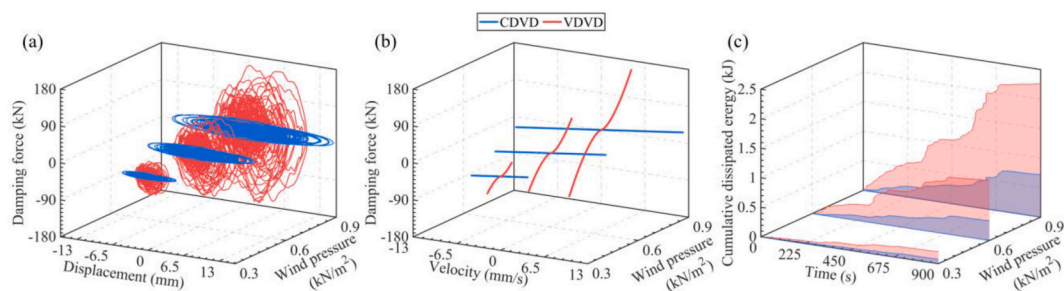


Fig. 28. Hysteretic curves and cumulative dissipated energy of CDVD and VDVD on the top floor when VDVD without a constant damping stage: (a) damping force-interlayer displacement curves; (b) damping force-interlayer velocity curves; (c) cumulative dissipated energy.

influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This project was funded by the National Natural Science Foundation of China [52178488]. The authors gratefully acknowledge financial support from the China Scholarship Council.

References

- [1] S.Y. Jeong, H. Alinejad, T.H.-K. Kang, Performance-based wind design of high-rise buildings using generated time-history wind loads, *J. Struct. Eng.* 147 (2021) 04021134, [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0003077](https://doi.org/10.1061/(ASCE)ST.1943-541X.0003077).

- [2] N. Satake, K. Suda, T. Arakawa, A. Sasaki, Y. Tamura, Damping evaluation using Full-scale data of buildings in Japan, *J. Struct. Eng.* 129 (2003) 470–477, [https://doi.org/10.1061/\(ASCE\)0733-9445\(2003\)129:4\(470\)](https://doi.org/10.1061/(ASCE)0733-9445(2003)129:4(470)).
- [3] K.D. Pitilakis, S.T. Karapetrou, S.D. Fotopoulou, Consideration of aging and SSI effects on seismic vulnerability assessment of RC buildings, *Bull Earthquake Eng* 12 (2014) 1755–1776, <https://doi.org/10.1007/s10518-013-9575-8>.
- [4] L. Wang, X.-Y. Fan, S.-G. Liang, J. Song, Z.-K. Wang, Improved expression for across-wind aerodynamic damping ratios of super high-rise buildings, *J. Wind Eng. Ind. Aerod.* 176 (2018) 263–272, <https://doi.org/10.1016/j.jweia.2018.04.001>.
- [5] Q. Wang, H. Tian, H. Qiao, N.D. Tiwari, Q. Wang, Wind-induced vibration control and parametric optimization of connected high-rise buildings with tuned liquid-column-damper-inerter, *Eng. Struct.* 226 (2021) 111352, <https://doi.org/10.1016/j.engstruct.2020.111352>.
- [6] Y. Zhang, T. Schauer, A. Bleicher, Optimized passive/semi-active vibration control using distributed-multiple tuned facade damping system in tall buildings, *J. Build. Eng.* 52 (2022) 104416, <https://doi.org/10.1016/j.jobe.2022.104416>.
- [7] M. Jafari, A. Alipour, Methodologies to mitigate wind-induced vibration of tall buildings: a state-of-the-art review, *J. Build. Eng.* 33 (2021) 101582, <https://doi.org/10.1016/j.jobe.2020.101582>.
- [8] J. Yu, Z. Duan, X. Zhang, J. Peng, Wind-induced vibration control of high-rise structures using compound damping cables, *Shock Vib.* 2021 (2021) e5537622, <https://doi.org/10.1155/2021/5537622>.
- [9] K. Zhou, J.-W. Zhang, Q.-S. Li, Control performance of active tuned mass damper for mitigating wind-induced vibrations of a 600-m-tall skyscraper, *J. Build. Eng.* 45 (2022) 103646, <https://doi.org/10.1016/j.jobe.2021.103646>.
- [10] S. Javadinasab Hormozabad, S. Zahrai, Innovative adaptive viscous damper to improve seismic control of structures, *J. Vib. Control* 25 (2019) 1833–1851, <https://doi.org/10.1177/1077546319841763>.
- [11] G. Pachideh, M. Gholhaki, M. Kafi, Experimental and numerical evaluation of an innovative diamond-scheme bracing system equipped with a yielding damper, *Steel Compos. Struct.* 36 (2020) 197–211, <https://doi.org/10.12989/scs.2020.36.2.197>.
- [12] G. Pachideh, M. Kafi, M. Gholhaki, Evaluation of cyclic performance of a novel bracing system equipped with a circular energy dissipater, *Structures* 28 (2020) 467–481, <https://doi.org/10.1016/j.istruc.2020.09.007>.
- [13] Y. Zhou, F. Shi, O.E. Ozbulut, H. Xu, D. Zi, Experimental characterization and analytical modeling of a large-capacity high-damping rubber damper, *Struct. Control Health Monit.* 25 (2018) e2183, <https://doi.org/10.1002/stc.2183>.
- [14] K. Zoltowski, A. Banas, M. Binczyk, P. Kalitowski, Control of the bridge span vibration with high coefficient passive damper. Theoretical consideration and application, *Eng. Struct.* 254 (2022) 113781, <https://doi.org/10.1016/j.engstruct.2021.113781>.
- [15] Z. He, F. Shi, Z. Lin, C. Zhang, Y. Zhou, F. Zhao, Experimental characterization on cyclic stability behavior of a high-damping viscoelastic damper, *Construct. Build. Mater.* 371 (2023) 130749, <https://doi.org/10.1016/j.conbuildmat.2023.130749>.
- [16] O. El-Khoury, H. Adeli, Recent advances on vibration control of structures under dynamic loading, *Arch Computat Methods Eng* 20 (2013) 353–360, <https://doi.org/10.1007/s11831-013-9088-2>.
- [17] K. Shiba, S. Mase, Y. Yabe, K. Tamura, Active/passive vibration control systems for tall buildings, *Smart Mater. Struct.* 7 (1998) 588, <https://doi.org/10.1088/0964-1726/7/5/003>.
- [18] W.Q. Zhu, M. Luo, L. Dong, Semi-active control of wind excited building structures using MR/ER dampers, *Probabilist. Eng. Mech.* 19 (2004) 279–285, <https://doi.org/10.1016/j.probengmech.2004.02.011>.
- [19] Y.X. Su, B.Y. Duan, Q. Wei, R.D. Nan, B. Peng, The wind-induced vibration control of feed supporting system for large spherical radio telescope using electrorheological damper, *Mechatronics* 13 (2003) 95–110, [https://doi.org/10.1016/S0957-4158\(01\)00042-3](https://doi.org/10.1016/S0957-4158(01)00042-3).
- [20] S.D. Bharti, S.M. Dumne, M.K. Shirmali, Seismic response analysis of adjacent buildings connected with MR dampers, *Eng. Struct.* 32 (2010) 2122–2133, <https://doi.org/10.1016/j.engstruct.2010.03.015>.
- [21] N.K. Hazaveh, G.W. Rodgers, J.G. Chase, S. Pampanin, Experimental test and validation of a direction- and displacement-dependent viscous damper, *J. Eng. Mech.* 143 (2017) 04017132, [https://doi.org/10.1061/\(ASCE\)EM.1943-7889.0001354](https://doi.org/10.1061/(ASCE)EM.1943-7889.0001354).
- [22] H. Zhao, B. Wang, G. Chen, Numerical study on a rotational hydraulic damper with variable damping coefficient, *Sci. Rep.* 11 (2021) 22515, <https://doi.org/10.1038/s41598-021-01859-2>.
- [23] W. Xu, Y. Wang, H. Guo, D. Du, S. Wang, Theoretical and experimental investigation on the seismic performance of a novel variable-damping viscous fluid damper, *J. Build. Eng.* 53 (2022) 104537, <https://doi.org/10.1016/j.jobe.2022.104537>.
- [24] S. Al-Subaihawi, C. Kolay, T. Marullo, J.M. Ricles, S.E. Quiel, Assessment of wind-induced vibration mitigation in a tall building with damped outriggers using real-time hybrid simulations, *Eng. Struct.* 205 (2020) 110044, <https://doi.org/10.1016/j.engstruct.2019.110044>.
- [25] J. Chen, X. Zeng, Y. Peng, Probabilistic analysis of wind-induced vibration mitigation of structures by fluid viscous dampers, *J. Sound Vib.* 409 (2017) 287–305, <https://doi.org/10.1016/j.jsv.2017.07.051>.
- [26] D. De Domenico, G. Ricciardi, Earthquake protection of structures with nonlinear viscous dampers optimized through an energy-based stochastic approach, *Eng. Struct.* 179 (2019) 523–539, <https://doi.org/10.1016/j.engstruct.2018.09.076>.
- [27] P.P. Diotallevi, L. Landi, A. Dellavalle, A methodology for the direct assessment of the damping ratio of structures equipped with nonlinear viscous dampers, *J. Earthq. Eng.* 16 (2012) 350–373, <https://doi.org/10.1080/13632469.2011.618521>.
- [28] E. Tubaldi, M. Barbato, A. Dall'Asta, Performance-based seismic risk assessment for buildings equipped with linear and nonlinear viscous dampers, *Eng. Struct.* 78 (2014) 90–99, <https://doi.org/10.1016/j.engstruct.2014.04.052>.
- [29] A. Ahmadi Nadooshan, M. Hemmat Esfe, M. Afrand, Evaluation of rheological behavior of 10W40 lubricant containing hybrid nano-material by measuring dynamic viscosity, *Phys. E Low-dimens. Syst. Nanostruct.* 92 (2017) 47–54, <https://doi.org/10.1016/j.physe.2017.05.011>.
- [30] E. Rigaud, J. Perret-Liaudet, M. Belin, L. Joly-Pottuz, J.-M. Martin, An original dynamic tribotest to discriminate friction and viscous damping, *Tribol. Int.* 43 (2010) 320–329, <https://doi.org/10.1016/j.triboint.2009.06.011>.
- [31] J.N. Yang, A.K. Agrawal, B. Samali, J.-C. Wu, Benchmark Problem for response control of wind-excited tall buildings, *J. Eng. Mech.* 130 (2004) 437–446, [https://doi.org/10.1061/\(ASCE\)0733-9399\(2004\)130:4\(437\)](https://doi.org/10.1061/(ASCE)0733-9399(2004)130:4(437)).
- [32] V.B. Patil, R.S. Jangid, Response of wind-excited benchmark building installed with dampers, *Struct. Des. Tall Special Build.* 20 (2011) 497–514, <https://doi.org/10.1002/tal.523>.
- [33] S.J. Suthar, V.B. Patil, R.S. Jangid, Optimization of MR dampers for wind-excited benchmark tall building, *Pract. Period. Struct. Des. Construct.* 27 (2022) 04022048, [https://doi.org/10.1061/\(ASCE\)SC.1943-5576.0000733](https://doi.org/10.1061/(ASCE)SC.1943-5576.0000733).
- [34] C. Zhang, Control force characteristics of different control strategies for the wind-excited 76-story benchmark building structure, *Adv. Struct. Eng.* 17 (2014) 543–559, <https://doi.org/10.1260/1369-4332.17.4.543>.
- [35] B. Samali, K.C.S. Kwok, G.S. Wood, J.N. Yang, Wind tunnel tests for wind-excited benchmark building, *J. Eng. Mech.* 130 (2004) 447–450, [https://doi.org/10.1061/\(ASCE\)0733-9399\(2004\)130:4\(447\)](https://doi.org/10.1061/(ASCE)0733-9399(2004)130:4(447)).
- [36] GB50009, *Load Code for the Design of Building Structures: GB50009-2012*, 2012.